

Formulas

CS Exe FM

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Chapter 11 - Introduction

$$1. \text{EVA} = (\text{Operating profit} - \text{Capital charge}) \\ = \text{NOPAT} - (\text{Cost of capital} \times \text{Capital})$$

$$2. \text{MVA} = \text{Value of firm} - \text{Capital invested} \\ = V - K$$

Chapter 12 - Time Value of Money

$$3. \text{PV} = \frac{\text{FV}}{(1+i)^n} \quad \begin{array}{l} i \rightarrow \text{Interest / DF} \\ n \rightarrow \text{no. of years} \end{array}$$

4. $FV = PV \times (1+i)^n$

5. In case of semi-annual compounding
 $i/2$ $n \times 2$

In case of quarterly compounding
 $i/4$ $n \times 4$

In case of monthly compounding
 $i/12$ $n \times 12$

6. PV of an annuity

$$P = PMT \times \frac{1 - (1+i)^{-n}}{i}$$

$P \rightarrow$ PV of annuity

$PMT \rightarrow$ monetary value of each annuity payment

7. FV of an annuity

$$P = PMT \times \frac{(1+i)^n}{i}$$

8. Rule of 72 $\rightarrow \frac{72}{\text{rate of interest}}$

9. Rule of 69 $\rightarrow 0.35 + \frac{69}{\text{rate of interest}}$

Chapter 13 - Capital Budgeting

10. PBP

Even period

Uneven period

Cash outflow

Even cash inflow

$$E + \frac{B}{C}$$

11. Discounted PBP

	(1)	(2)	(1x2)	
Year	Cash Flow	DF	PV	Cum. PV

$$E + \frac{B}{C}$$

12. NPV = PVCF - PVCO

	(1)	(2)	(1x2)
Year	Cash Flow	DF	PV

$$13. \quad PK = \frac{PVC \cdot K}{PVC0}$$

$$14. \quad ROR/ARR = \frac{\text{Avg. annual CF} - \text{Annual Dep.} \times 100}{\text{Avg. investment}}$$

$$\text{Avg. investment} = \frac{\text{Initial invt.} + \text{Scrap}}{2}$$

$$\text{Annual Dep.} = \frac{\text{Cost of machine} - \text{Scrap}}{\text{life of machine}}$$

$$15. \quad LRR = LR + \frac{NPV_{LR}}{NPV_{LR} - NPV_{HR}} \times (HR - LR)$$

LR $\rightarrow$ Lower rate

HR $\rightarrow$ Higher rate

$$16. \quad RADR = RF + \text{Risk premium rate}$$

$$17. \quad \text{Certainty equivalent Co-efficient}$$

$$= \frac{\text{Riskless cash flow}}{\text{Risky cash flow}}$$

or

$$CF \times CE = \frac{\text{Certain}}{CF} \times DF = PV$$

18. Standard deviation

Step 1 : Calculate average $\left(\frac{\sum x}{n} \right) = \bar{x}$

Step 2 : Calculate deviation $(x - \bar{x})$

Step 3 : Square of deviation

Step 3 : $\sum (\text{Square of deviation} \times \text{Probability})$

Step 4 : $\sqrt{\text{step 3 value}}$

Chapter 14

19. Cost of debt

Redeemable

Irredeemable

$$\frac{R(1-t)}{\left(\frac{RV + NP}{2} \right)} + \frac{(RV - NP)}{n} \times 100$$

$$\frac{R(1-t)}{NP} \times 100$$

RV $\rightarrow$ Redeemable value / maturity value
 NP $\rightarrow$ Net proceeds
 t $\rightarrow$ tax rate

20. Cost of preference share

↓
Redeemable
↓

↓
Irredeemable
↓

$$\frac{PD + \left(\frac{RV - NP}{n} \right) \times 100}{\left(\frac{RV + NP}{2} \right)}$$

$$\frac{PD}{NP} \times 100$$

21. Cost before tax $\rightarrow$ $\frac{\text{Cost after tax}}{(1 - t)}$

22. Cost of equity

a) CAPM $\rightarrow R_f + \beta(R_m - R_f)$

R_f = Risk free rate

R_m = Market risk

β = Beta

b) Dividend yield model

$$P_0 = \frac{DPS}{MPS} \times 100$$

c) Earning yield method

$$P_0 = \frac{EPS}{MPS} \times 100$$

d) Growth

$$P_0 = \frac{\text{Dividend}}{K_E - g}$$

23. Cost of retained earning (K_{RE})

a) $K_{RE} = K_E \times (1 - \text{Personal income tax})$

b) $K_{RE} = \frac{DPS (1 - PRT) (1 - \text{brokerage})}{MP (1 - \text{Capital gain})}$

24. $WACC = E \left[\frac{(K_d \times W_d) + (K_e \times W_e) + (K_p \times W_p) + (K_{RE} \times W_{RE})}{1} \right]$

Chapter 15 : Capital Structure

25. NZ Approach

$$K_0 = K_d \times \frac{D}{(D+E)} + K_e \times \frac{E}{(D+E)}$$

$$V_f = \frac{EBIT}{K_o} \quad V_e = \frac{\text{Net Income}}{K_e}$$

$$V_d = \frac{\text{Int.}}{K_d}$$

26. NOI Approach

$$K_e = K_o + (K_o - K_d) \times \frac{D}{E}$$

$$V_f = \frac{\text{Net operating income}}{\text{WACC or } K_o}$$

$$V_e = \frac{\text{Net income after interest i.e. } (EBIT - \text{Interest})}{K_e}$$

27. Leverage

$$DOL \rightarrow \frac{\text{Contribution}}{EBIT}$$

$$DFL \rightarrow \frac{EBIT}{EBT}$$

$$DCL \rightarrow DOL \times DFL \quad \text{or} \quad \frac{\text{Contribution}}{EBT}$$

28. Financial Break even point

$$= \text{Interest} + \frac{\text{Preference dividend}}{(1-t)}$$

29. Indifference point

$$\frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of shares in plan A}} - \text{PD}(1+t) = \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of shares in plan B}} - \text{PD}(1+t)$$

30. Leverage in terms of percentage

$$\text{DCL} = \frac{\% \text{ change in EPS}}{\% \text{ change in sales}}$$

$$\text{DOL} = \frac{\% \text{ change in EBIT}}{\% \text{ change in sales}}$$

$$\text{DFL} = \frac{\% \text{ change in EPS}}{\% \text{ change in EBIT}}$$

31. Hamada Equation

Beta Levered company

$$= \beta_{\text{unlevered co.}} \left[1 + (1-t) \frac{D}{E} \right]$$

32. Walter model

$$P_0 = \frac{DPS + \left(\frac{R}{K_e} \right) (EPS - DPS)}{K_e}$$

DPS $\rightarrow$ Dividend/share

EPS $\rightarrow$ Earning/share

33. Gordon model

$$P_0 = \frac{D_0(1+g)}{K_e - g}$$

34. Working Capital

$$\text{Net WC} = \text{Current Asset (C.A.)} - \text{Current liability (C.L.)}$$

$$C.A. = RM + WIP + FG + Debtors + Cash$$

$$C.L. = \text{lag in payment of wages} + \text{creditors} + \text{etc}$$

* Units $\times$ Cost $\times$ Time period

35. Operating cycle

$$= \text{Rm} + \text{WIP} + \text{FG} + \text{Debtors} - \text{creditors}$$

$$\text{Rm} \rightarrow \frac{\text{Avg. inventory of Rm}}{\text{Avg. consumption of Rm}}$$

$$\text{WIP} \rightarrow \frac{\text{Avg. WIP}}{\text{Avg. cost of production}}$$

$$\text{FG} \rightarrow \frac{\text{Avg. inventory of FG}}{\text{Avg. cost of sales}}$$

$$\text{Debtors} \rightarrow \frac{\text{Avg. Debtors}}{\text{Avg. Sales}}$$

$$\text{Creditors} \rightarrow \frac{\text{Avg. Creditors}}{\text{Avg. credit purchases}}$$

36. Re order level = maximum consumption $\times$ maximum re-order period

37. Minimum Stock level = Re-order level - (Normal consumption $\times$ Normal re-order period)

38. Maximum stock level

$$= \text{Re-order level} + \text{Re-order quantity} \\ - (\text{min. consumption} \times \text{min. re-order period})$$

39. Danger level

$$= \text{Average consumption} \times \text{maximum re-ordering period for emergency purchase}$$

40. Average stock level

$$= \frac{\text{min. stock} + \text{max. stock}}{2}$$

$$41. \text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}}$$

A $\rightarrow$ Annual Demand

O $\rightarrow$ Ordering cost

C $\rightarrow$ Carrying cost

42. Calculation of Total cost in EOQ

$$= (\text{Purchase price} \times \text{Demand}) + \left(\frac{\text{EOQ}}{2} \times \text{Carrying cost} \right) + \left(\frac{\text{Demand} \times \text{order cost}}{\text{EOQ}} \right)$$

43. Relative Strength Index

$$RSI = 100 - \frac{100}{1 + RS}$$

$$RS = \frac{\text{Avg. gain per day}}{\text{Avg. loss per day}}$$

44. Aroon

$$\text{Aroon up} = \frac{25 - \text{Period since 25 period high}}{25} \times 100$$

$$\text{Aroon down} = \frac{25 - \text{Period since 25 period low}}{25} \times 100$$

45. Price rate of change = $\frac{\text{Closing price} - \text{opening price}}{\text{opening price}} \times 100$

46. Holding period return (HPR)

$$= \left[\frac{\text{Income} + \text{closing value} - \text{opening value}}{\text{opening value}} \right]^{1/n} - 1$$

47. Marginal costing

a) $\text{Contribution} = \text{Sales} - \text{Variable cost}$

b) $\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}}$

or

$$\frac{\text{change in profit}}{\text{change in sales}}$$

c) Break even point

Number

$\bar{x}$

Fixed cost

Fixed cost

contribution/unit

P/V Ratio

d) $\text{Margin of safety} = \text{Actual sales} - \text{Break even sales}$

e) $\text{Profit} = \text{margin of safety} \times \text{P/V Ratio}$